



Fair, Accountable, Transparent Machine Learning (FAccT ML)

ZG517



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Session 3
Date – 7th Dec 2024
Time – 10:40 AM to 12:40 PM

These slides are prepared by the instructor, with grateful acknowledgement of many others who made their course materials freely available online.

Session Content



- FICO Case Study
- Fair Methods for ML
 - Preprocessing Methods
 - Reweighting
 - Resampling
 - ✓ Universal
 - ✓ Preferential
 - Optimized Preprocessing
 - In-processing Methods
 - Prejudice Preserving Regularization
 - Adversarial Learning
 - Post-processing Methods

Fair ML Methods



- **Pre-processing Methods**

- Transform data before ML models learn
- e.g., Reweighting, Resampling
- Additional Optimization

- **In-processing Methods**

- Constrain ML models while they learn, e.g.,
 - Prejudice Removing Regularizer
 - Adversarial Learning

- **Post-processing Methods**

- Make predictions from a black-box ML model fair in the post-processing stage
- e.g., Learning to Defer

Fair Data Manipulation



- Biased Data
 - The presence of data that belongs to the underrepresented groups leads to data biases
 - One of the main sources of ML discriminations
- Data Debiasing
 - Adjust the distribution of the data to meet fairness criteria
 - Increase/Decrease samples based on criteria
- Reweighting
 - Adjust the importance of each sample in the loss function during training
- Resampling
 - Adjust the proportion of samples for each group

Basic Data Manipulation Techniques

- Reweighing
- Universal Sampling
- Preferential Sampling

Biased Data



Observed: $M = 10$, $F = 4$



Expected: $M = 7$, $F = 7$

Expected Distribution of Fair Data

- Expected Data Distribution

$$P(Y) = P(Y|A = 1) = P(Y|A = 0)$$

which leads to $Y \perp\!\!\!\perp A$

- Recall Demographic Parity

$$P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$$

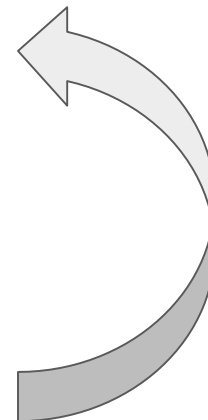
Expected Distribution of Fair Data

- The Expected Joint Distribution Under $Y \perp\!\!\!\perp A$

$$\begin{aligned} P_{\text{exp}}(Y = y, A = a) &= P(Y = y) \cdot P(A = a) \\ &= \frac{|\{x \in \mathcal{D} | x_Y = y\}|}{|\mathcal{D}|} \cdot \frac{|\{x \in \mathcal{D} | x_A = a\}|}{|\mathcal{D}|} \end{aligned}$$

- Our Observed Joint Distribution

$$P_{\text{obs}}(Y = y, A = a) = \frac{|\{x \in \mathcal{D} | x_Y = y, x_A = a\}|}{|\mathcal{D}|}$$



Transform Data to
Expected Distribution

[Kamiran et al, 2012](#)

Reweighting

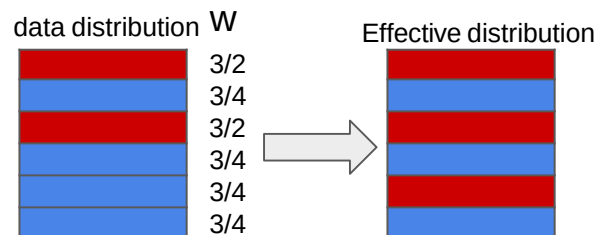


- Sample Weight for x
 - Goal: adjust our data to a distribution that leads to $Y \perp\!\!\!\perp A$, or Demographic Parity
 - $W(x) = 1$, we have achieved $Y \perp\!\!\!\perp A$ and Demographic Parity
 - $W(x) > 1$, increase the weight of sample x in training
 - $W(x) < 1$, decrease the weight of sample x in training

$$W(x) = \frac{P_{exp}(Y = x_y, A = x_a)}{P_{obs}(Y = x_y, A = x_a)}$$

- Reweighting Loss Function

$$\mathcal{L} = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} W(x) \cdot \mathcal{L}(\hat{Y}, x_Y)$$



Practice Question

- Calculate $W(x_3)$, $A = \{\text{Sex}\}$, $Y = \{\text{Class}\}$

Sex	Highest degree	Job type	Class
M	H. school	Board	+
M	Univ.	Board	+
M	H. school	Board	+
M	H. school	Healthcare	+
M	Univ.	Healthcare	—
F	Univ.	Education	—
F	H. school	Education	—
F	None	Healthcare	+
F	Univ.	Education	—
F	H. school	Board	+

[Kamiran et al, 2012](#)

Practice Question

- $W(x_3)$
 - $A_3 = M$
 - $Y_3 = +$
- Expected Distribution
 - $P(A = M) = 0.5$
 - $P(Y = +) = 0.6$
 - $P_{\text{exp}}(A = M, Y = +) = 0.3$
- Observed Distribution
 - $P_{\text{obs}}(A = M, Y = +) = 0.4$
- Sample Weight
 - $W(x_3) = 0.3/0.4 = 0.75$

$A = \{\text{Sex}\}, Y = \{\text{Class}\}$

Sex	Highest degree	Job type	Class
M	H. school	Board	+
M	Univ.	Board	+
M	H. school	Board	+
M	H. school	Healthcare	+
M	Univ.	Healthcare	—
F	Univ.	Education	—
F	H. school	Education	—
F	None	Healthcare	+
F	Univ.	Education	—
F	H. school	Board	+

[Kamiran et al, 2012](#)

Quiz



- Calculate $W(x_6)$, $A = \{\text{Sex}\}$, $Y = \{\text{Class}\}$

Sex	Highest degree	Job type	Class
M	H. school	Board	+
M	Univ.	Board	+
M	H. school	Board	+
M	H. school	Healthcare	+
M	Univ.	Healthcare	—
F	Univ.	Education	—
F	H. school	Education	—
F	None	Healthcare	+
F	Univ.	Education	—
F	H. school	Board	+

[Kamiran et al, 2012](#)

Answer



- $W(x_6)$
 - $A_6 = F$
 - $Y_6 = -$
- Expected Distribution
 - $P(A = F) = 0.5$
 - $P(Y = -) = 0.4$
 - $P_{\text{exp}}(A = F, Y = -) = 0.2$
- Observed Distribution
 - $P_{\text{obs}}(A = F, Y = -) = 0.3$
- Sample Weight
 - $W(x_6) = 0.2/0.3 = 0.67$

Sex	Highest degree	Job type	Class
M	H. school	Board	+
M	Univ.	Board	+
M	H. school	Board	+
M	H. school	Healthcare	+
M	Univ.	Healthcare	-
F	Univ.	Education	-
F	H. school	Education	-
F	None	Healthcare	+
F	Univ.	Education	-
F	H. school	Board	+

Homework



- Calculate $W(x_1) \dots W(x_{10})$
- Put $W(x_i)$ into the loss

$$\mathcal{L} = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} W(x) \cdot \mathcal{L}(\hat{Y}, x_Y)$$

Can we achieve data
pre-processing for fairness without
changing the training objective?

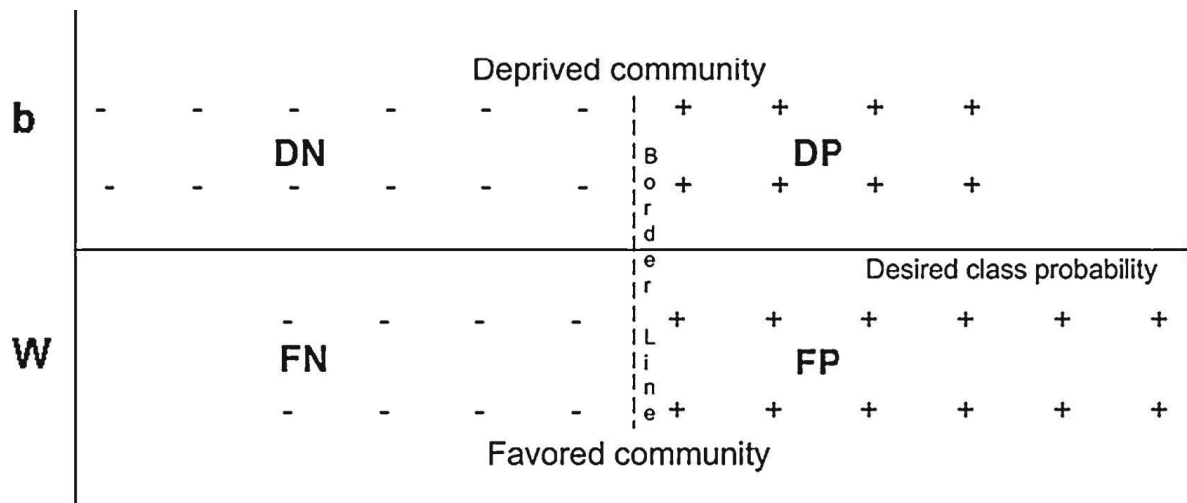
$A = \{\text{Sex}\}, Y = \{\text{Class}\}$

Sex	Highest degree	Job type	Class
M	H. school	Board	+
M	Univ.	Board	+
M	H. school	Board	+
M	H. school	Healthcare	+
M	Univ.	Healthcare	—
F	Univ.	Education	—
F	H. school	Education	—
F	None	Healthcare	+
F	Univ.	Education	—
F	H. school	Board	+

Resampling



- Resample the Dataset Based on the Expected Joint Probability



[Kamiran et al, 2012](#)

Expected Number of Samples

- Expected Number of Samples for the Category (y, a)

$$N_{exp}(y, a) = P_{exp}(y, a) \cdot |\mathcal{D}|$$

- Also Note

$$\sum_{y,a} N_{exp} = \sum_{y,a} P_{exp}(y, a) \cdot |\mathcal{D}| = |\mathcal{D}|$$

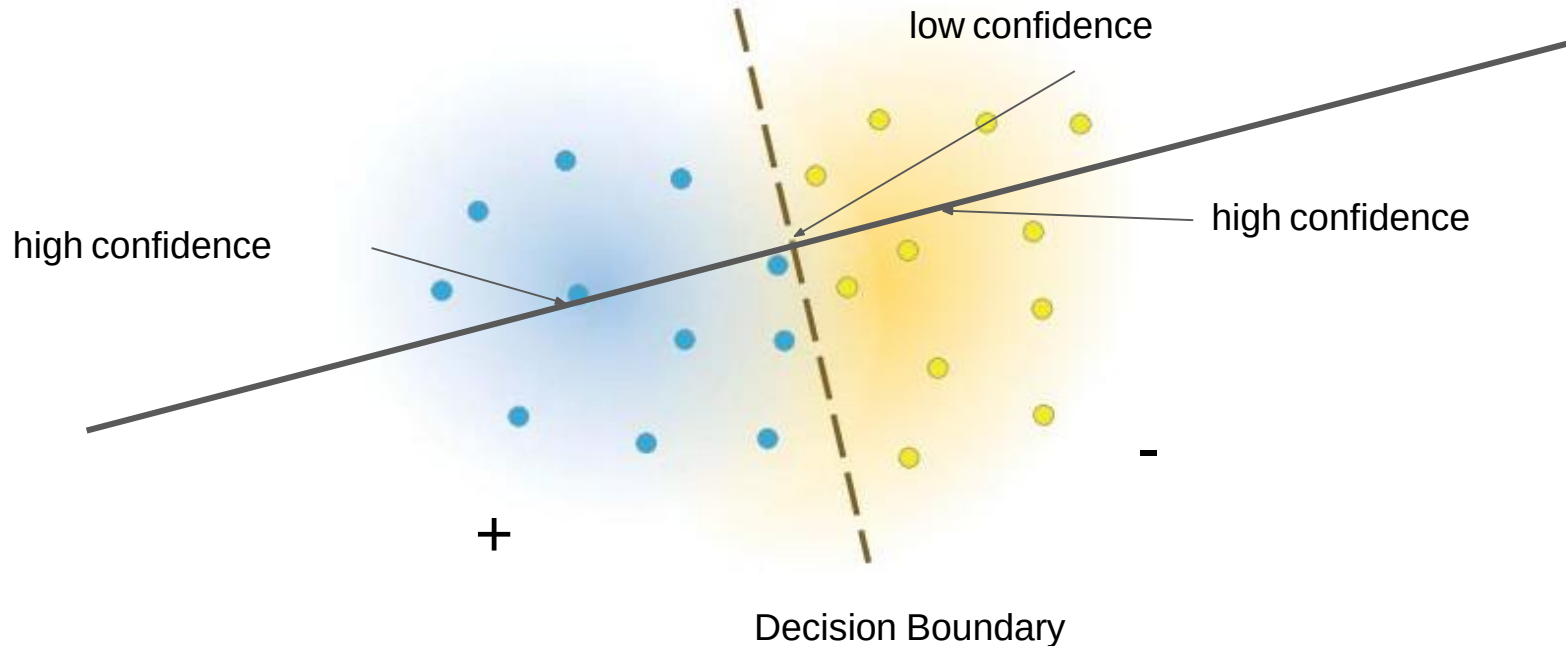
Universal Resampling (US)



- Resampling Based on the Expected Probabilities to Meet Demographic Parity
 - DP (Deprived community with Positive class labels)
 - draw $N_{\text{exp}}(D, P)$ samples uniformly from DP
 - DN (Deprived community with Negative class labels)
 - draw $N_{\text{exp}}(D, N)$ samples uniformly from DN
 - FP (Favored community with Positive class labels)
 - draw $N_{\text{exp}}(F, P)$ samples uniformly from FP
 - FN (Favored community with Negative class labels)
 - draw $N_{\text{exp}}(F, N)$ samples uniformly from FN

Preferential Sampling (PS)

- Sample More Data When Confidence of the Predictor Is Low



[Kamiran et al, 2012](#)

Bias Measures

- Measure prediction biases by comparing the favorable outcomes given to group 1 with that to group 0

$$Bias(\hat{Y}) = P(\hat{Y} = 1|A = 1) - P(\hat{Y} = 1|A = 0)$$

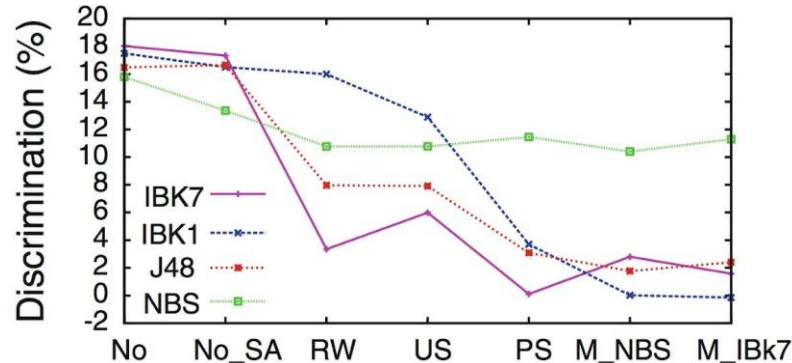
Demographic Parity

$$P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$$

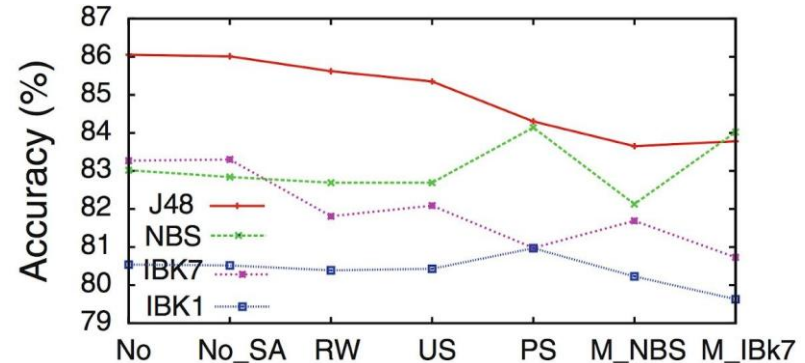
Adult Income Dataset



No - No pre-processing, No-SA - No Sex Attribute, RW - Reweighting
US - Universal Sampling, PS - Preferential Sampling
M_* - “massaged” input data (required self-study - refer to the paper)



J48 - decision tree
NBS - Naive Bayes



IBK1- 1 nearest neighbor
IBK7 -7 nearest neighbor

[Kamiran et al, 2012](#)

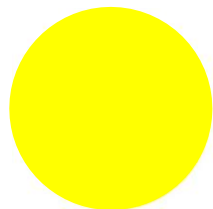
Continuous Data?



$$N_{exp}(y, a) = P_{exp}(y, a) \cdot |\mathcal{D}|$$

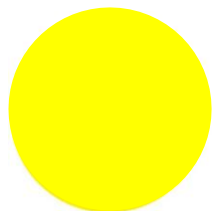
$$W(x) = \frac{P_{exp}(Y = x_y, A = x_a)}{P_{obs}(Y = x_y, A = x_a)}$$

Individual Fairness



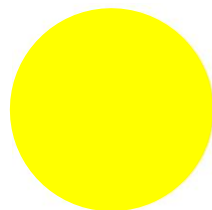
Income = \$50k
Credit Score = 690

Accepted



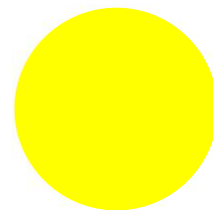
Income = \$43k
Credit Score = 650

Accepted



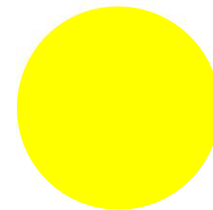
Income = \$50k
Credit Score = 690

Denied
???



Income = \$70k
Credit Score = 740

Accepted



Income = \$100k
Credit Score = 750

Accepted

group 1

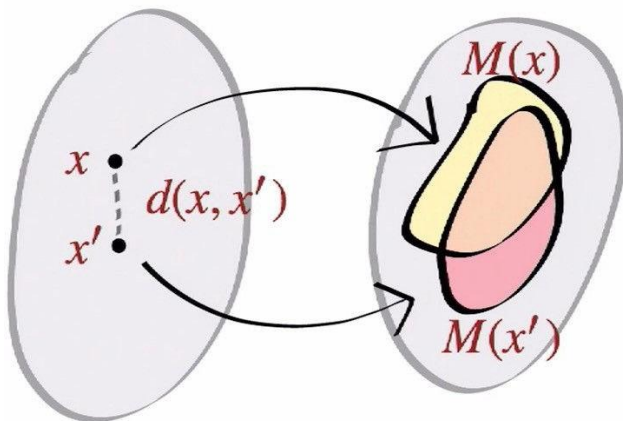
group 2

Individual Fairness



- A predictor M achieves individual fairness under a distance metric d iff
 - Similar Samples are treated similarly, in other words

$$M(x_i) \approx M(x_j) | d(x_i, x_j) \approx 0$$



Individual Fairness



Individual

$$M(x_i) \approx M(x_j) | d(x_i, x_j) \approx 0$$



Income = \$19k
Credit Score = 690



Income = \$23k
Credit Score = 720



Income = \$60k
Credit Score = 800

Group 1



Income = \$20k
Credit Score = 680



Income = \$27k
Credit Score = 700



Income = \$65k
Credit Score = 810

Group 2

Fairness Criteria



Individual Treatment	Group Treatment
Fairness Through Unawareness Excludes Sensitive Information A from the predictor	Demographic Parity $P(\hat{Y} = 1 A = 1) = P(\hat{Y} = 1 A = 0)$
Individual Fairness $M(x_i) \approx M(x_j) d(x_i, x_j) \approx 0$	Equal Opportunity/Odds $P(\hat{Y} = 1 A = 0, Y = 1) = P(\hat{Y} = 1 A = 1, Y = 1)$ $P(\hat{Y} = 1 A = 0, Y) = P(\hat{Y} = 1 A = 1, Y)$

Optimized Pre-Processing for Fairness



- Can We Automate the Resampling Process?

- Turn the manual process into an optimization based approach
- Include more criteria than Demographic Fairness
- Allow transformations of data

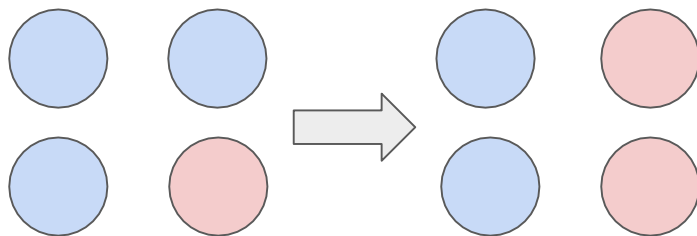
- Optimized Pre-Processing

- Given sensitive feature D , learn a probabilistic mapping $p_{\hat{X}, \hat{Y} | X, Y, D}$ that satisfies

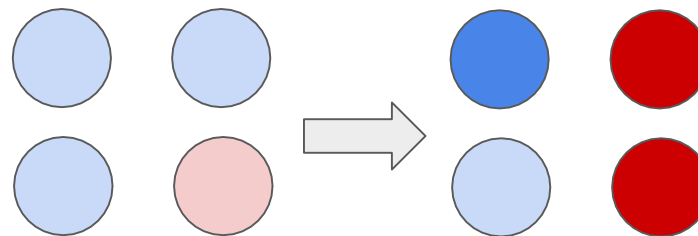
$$\{(D_i, X_i, Y_i)\}_{i=1}^n \xrightarrow{p_{\hat{X}, \hat{Y} | X, Y, D}} \{(D_i, \hat{X}_i, \hat{Y}_i)\}_{i=1}^n$$

[Calmon et al, 2017](#)

Resampling and Transforming



Resampling

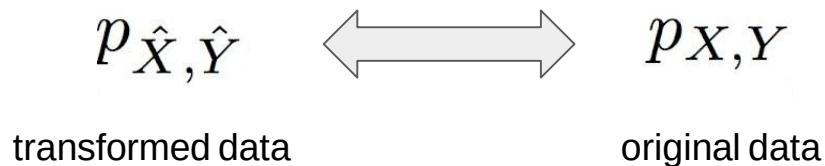


Transforming

Constraint 1: Utility Preservations



- A Utility Function to Preserve the Joint Probability
 - e.g. KL Divergence



Constraint 2: Discrimination Control



- Constrain the dependency of the target variable y given sensitive feature d to match target $p_{Y_T}(y)$
 - J - distance measure $J(p, q) = \left| \frac{p}{q} - 1 \right|$
 - $\epsilon_{y,d}$ - a small number used as our tolerance

$$J \left(p_{\hat{Y}|D}(y|d), p_{Y_T}(y) \right) \leq \epsilon_{y,d} \quad \forall d \in \mathcal{D}, y \in \{0, 1\}$$

When $p_{\hat{Y}|D}(y|d) = p_{Y_T}(y)$, we achieve Demographic Parity

Constraint 3: Distortion Control



- An Implementation of the Individual Fairness

$$M(x_i) \approx M(x_j) | d(x_i, x_j) \approx 0$$

- The Mapped Sample $\hat{X}, \hat{Y}$ Has to Stay Close to the Original Sample x, y
 - $c_{d,x,y}$ - tolerance
 - δ - a similarity function
 - 1 - very different
 - 0 - very similar

$$\Pr \left(\delta((x, y), (\hat{X}, \hat{Y})) = 1 \mid D = d, X = x, Y = y \right) \leq c_{d,x,y}$$

Putting Things Together



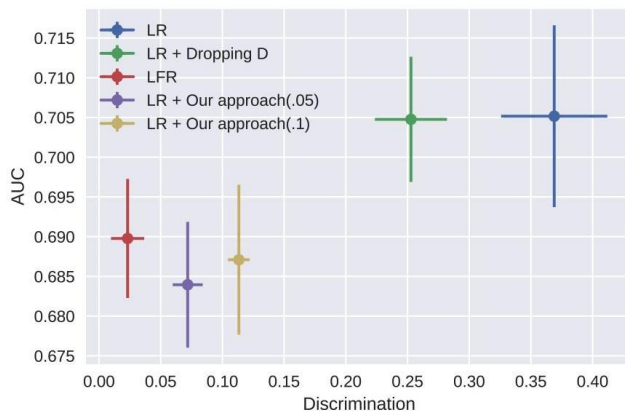
$$\begin{aligned} \min_{p_{\hat{X}, \hat{Y} | X, Y, D}} \quad & \Delta \left(p_{\hat{X}, \hat{Y}}, p_{X, Y} \right) \\ \text{s.t.} \quad & J \left(p_{\hat{Y} | D}(y|d), p_{Y_T}(y) \right) \leq \epsilon_{y,d} \text{ and} \\ & \mathbb{E} \left[\delta((x, y), (\hat{X}, \hat{Y})) \mid D = d, X = x, Y = y \right] \leq c_{d,x,y} \end{aligned}$$

Utility

Discrimination control
group fairness

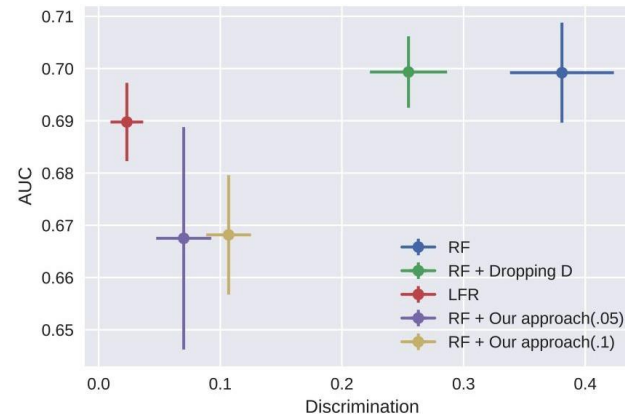
Distortion Control
Individual fairness

Results on COMPAS dataset



Logistic Regression

LFR - Learning Fair Representations (benchmark)



Random Forest

Fair ML Methods



- Pre-processing Methods
 - Transform data before ML models learn
 - e.g., Reweighting, Resampling
- In-processing Methods
 - Constrain ML models while they learn, e.g.,
 - **Prejudice Removing Regularizer**
 - **Adversarial Learning**
- Post-processing Methods
 - Make predictions from a black-box ML model fair in the post-processing stage
 - e.g., Learning to Defer

Prejudice Remover Regularizer

Quantified causes of unfairness



Prejudice

- Unfairness rooted in the dataset

Underestimation

- Model unfairness because the model is not fully converged

Negative Legacy

- Unfairness due to sampling biases

Training Objective

$$- \underbrace{\mathcal{L}(\mathcal{D}; \Theta)}_{\text{Loss of the Model}} + \underbrace{\eta R(\mathcal{D}, \Theta)}_{\text{Fairness Regularizer}} + \underbrace{\frac{\lambda}{2} \|\Theta\|_2^2}_{\text{L2 Regularizer}}$$

[Kamishima et al, 2012](#)

Prejudice Index (PI)

Recall that Indirect Discrimination Happens When



Prediction is not directly conditioned on sensitive variables S

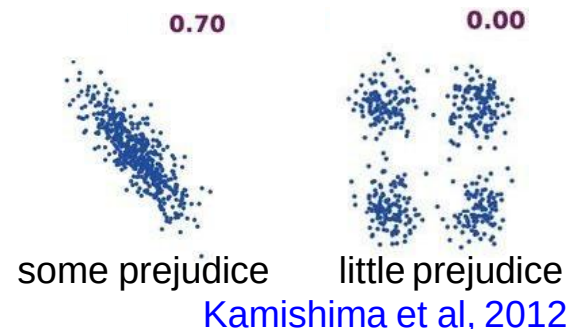
Prediction is indirectly conditioned on S by a variable O that is dependent on S

Prejudice Index (PI)

- Measures the degree of indirect discrimination based on mutual information

$$PI = \sum_{(y,s) \in \mathcal{D}} \hat{P}_r[y, s] \ln \frac{\hat{P}_r[y, s]}{\hat{P}_r[y] \hat{P}_r[s]}$$

↑
prediction model



[Kamishima et al, 2012](#)

Normalized Prejudice Index (NPI)

- Prejudice Index (PI)
 - Measures the degree of indirect discrimination based on mutual information
 - Ranges in $[0, +\infty)$

$$PI = \sum_{(y,s) \in \mathcal{D}} \hat{Pr}[y, s] \ln \frac{\hat{Pr}[y, s]}{\hat{Pr}[y] \hat{Pr}[s]}$$

- Normalized Prejudice Index (NPI)
 - Normalize PI by the entropy of Y and S
 - Ranges in $[0, 1]$

$$NPI = PI / (\sqrt{H(Y)H(S)})$$

Optimizing PI



- Learning PI

$$PI = \sum_{Y,S} \hat{Pr}[Y, S] \ln \frac{\hat{Pr}[Y, S]}{\hat{Pr}[S] \hat{Pr}[Y]}$$

Optimizing PI



- Learning PI

Expands $\Pr(Y, S)$ into $\sum_x \Pr(X, Y, S)$

$$PI = \sum_{Y,S} \hat{Pr}[Y, S] \ln \frac{\hat{Pr}[Y, S]}{\hat{Pr}[S] \hat{Pr}[Y]} = \sum_{X,S} \tilde{Pr}[X, S] \sum_Y \mathcal{M}[Y|X, S; \Theta] \ln \frac{\hat{Pr}[Y, S]}{\hat{Pr}[S] \hat{Pr}[Y]}.$$

double summations triple summations Prediction Model

- Using Logistic Regression Model as the Prediction Model

$$\mathcal{M}[y|\mathbf{x}, s; \Theta] = y\sigma(\mathbf{x}^\top \mathbf{w}_s) + (1 - y)(1 - \sigma(\mathbf{x}^\top \mathbf{w}_s))$$

Optimizing PI



- Learning PI

$$\begin{aligned} \text{PI} &= \sum_{Y,S} \hat{\text{Pr}}[Y, S] \ln \frac{\hat{\text{Pr}}[Y, S]}{\hat{\text{Pr}}[S] \hat{\text{Pr}}[Y]} = \sum_{X,S} \tilde{\text{Pr}}[X, S] \sum_Y \mathcal{M}[Y|X, S; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[Y, S]}{\hat{\text{Pr}}[S] \hat{\text{Pr}}[Y]} \\ &= \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}. \end{aligned}$$

- Using Logistic Regression Model as the Prediction Model

difficult to evaluate

$$\mathcal{M}[y|\mathbf{x}, s; \boldsymbol{\Theta}] = y\sigma(\mathbf{x}^\top \mathbf{w}_s) + (1 - y)(1 - \sigma(\mathbf{x}^\top \mathbf{w}_s))$$

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \Theta] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

[Kamishima et al, 2012](#)

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

$$\hat{\text{Pr}}[y|s] = \int_{\text{dom}(X)} \text{Pr}^*[X|s] \mathcal{M}[y|X, s; \boldsymbol{\Theta}] dX$$

Integrals Are Difficult to Evaluate

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

$$\begin{aligned} \hat{\text{Pr}}[y|s] &= \int_{\text{dom}(X)} \text{Pr}^*[X|s] \mathcal{M}[y|X, s; \boldsymbol{\Theta}] dX \\ &\approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|} \end{aligned}$$

Approximating integrals by sample means

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

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Approximating integrals by sample means

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

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Approximating integrals by sample means

$$\hat{\text{Pr}}[y] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}]}{|\mathcal{D}|}$$

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

$$\hat{\text{Pr}}[y|s] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i=s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|} \quad \hat{\text{Pr}}[y] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}]}{|\mathcal{D}|}$$

[Kamishima et al, 2012](#)

Putting Things Together



Optimization Target

$$- \mathcal{L}(\mathcal{D}; \Theta) + \eta R(\mathcal{D}, \Theta) + \frac{\lambda}{2} \|\Theta\|_2^2$$

Loss of the Model

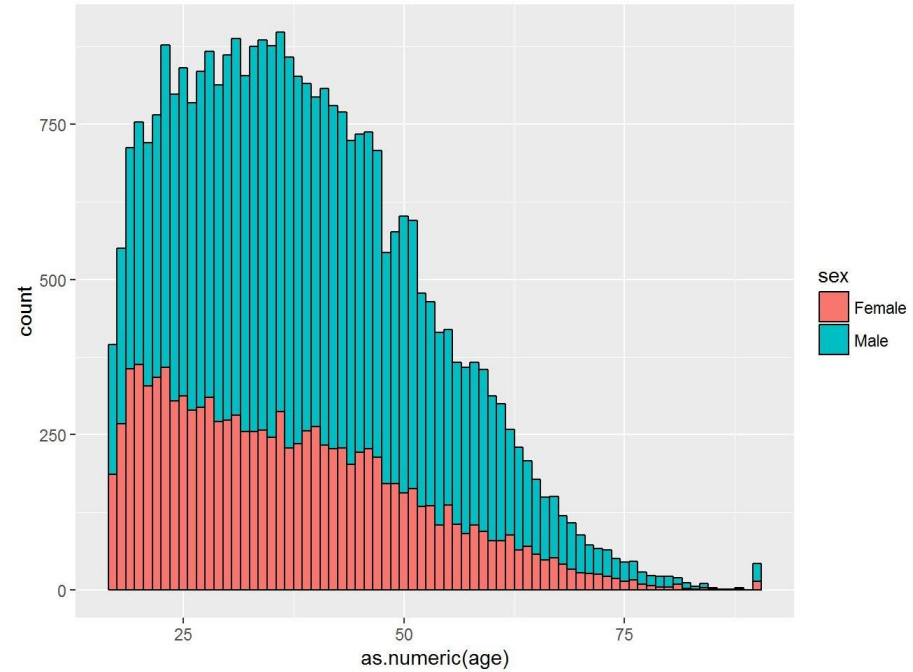
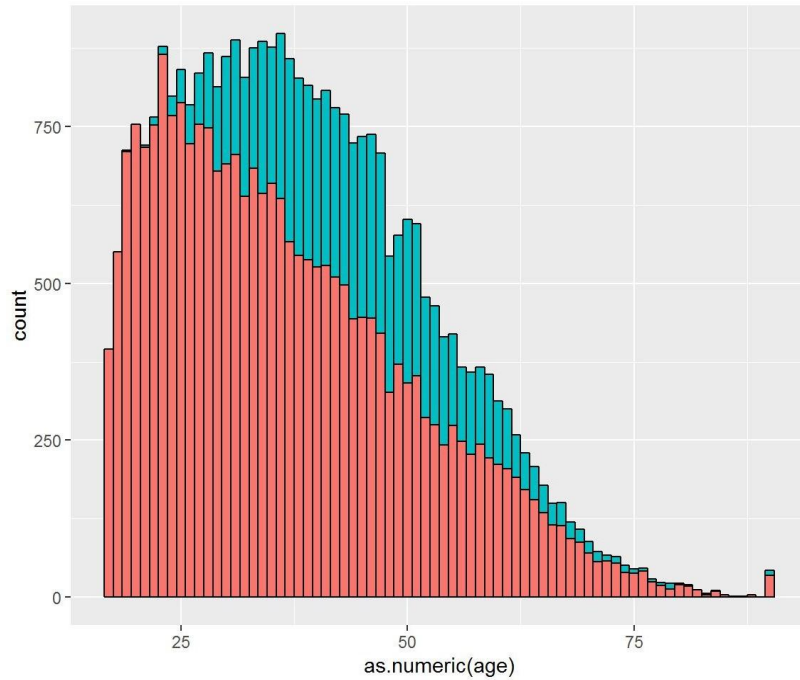
Fairness Regularizer

L2 Regularizer

- Fairness Regularizer

$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \Theta] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

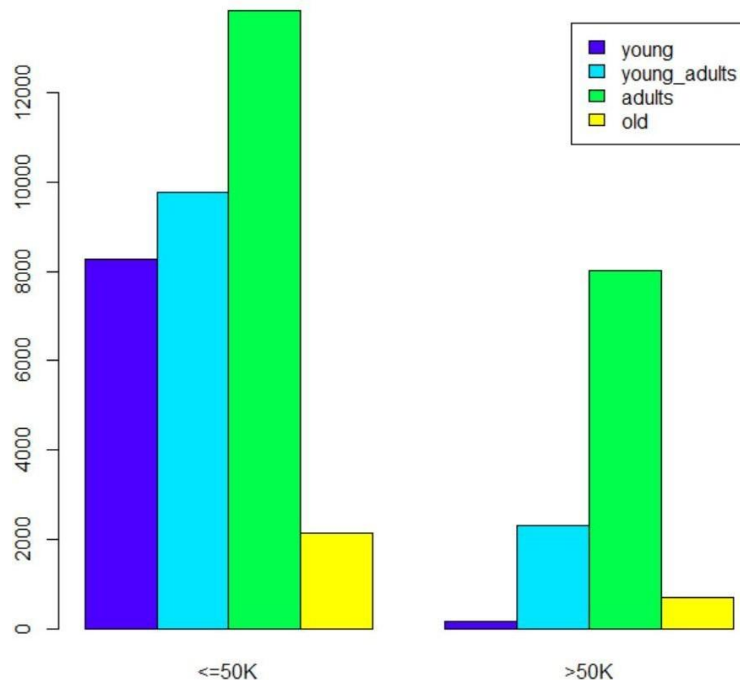
Adult Income Dataset (Kohavi 1996)



Adult Income Dataset ([Kohavi 1996](#))



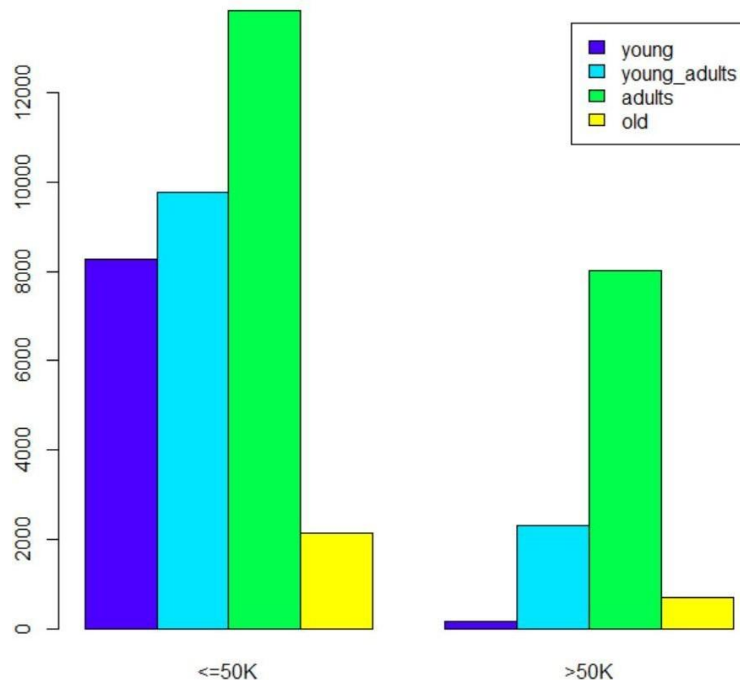
- Predict Whether Income Exceeds \$50K/yr Based on Census Data



Adult Income Dataset ([Kohavi 1996](#))



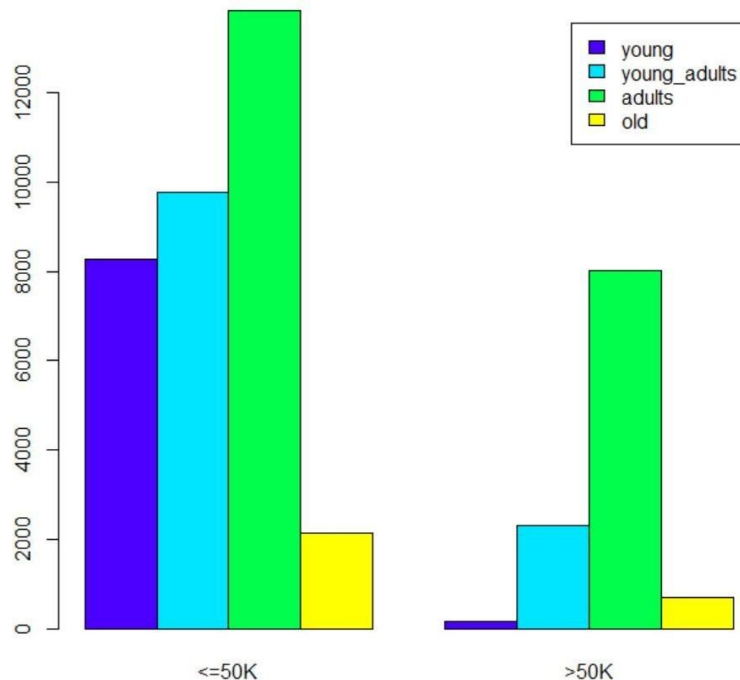
- Predict Whether Income Exceeds \$50K/yr Based on Census Data



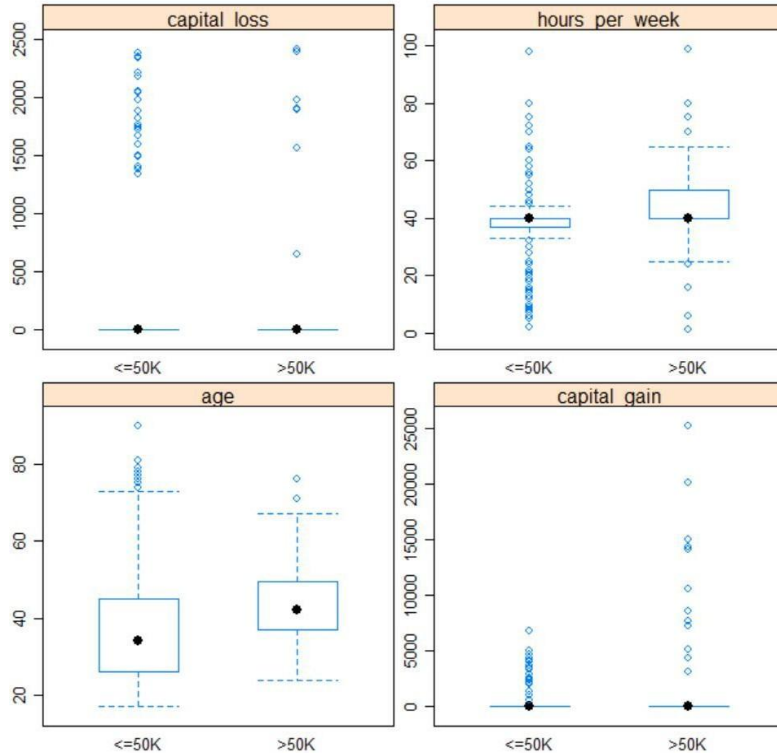
Adult Income Dataset ([Kohavi 1996](#))



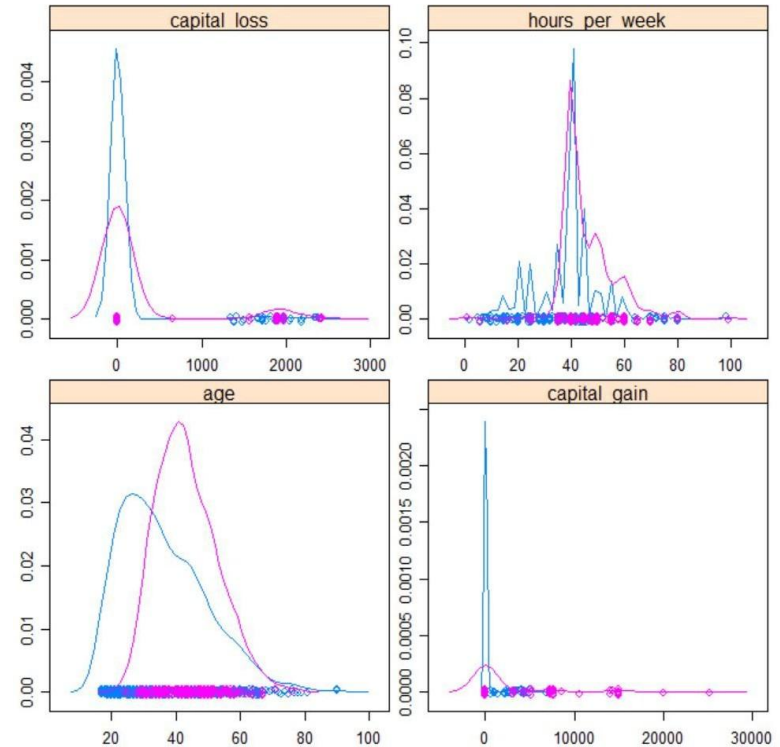
- Predict Whether Income Exceeds \$50K/yr Based on Census Data



Adult Income Dataset ([Kohavi 1996](#))

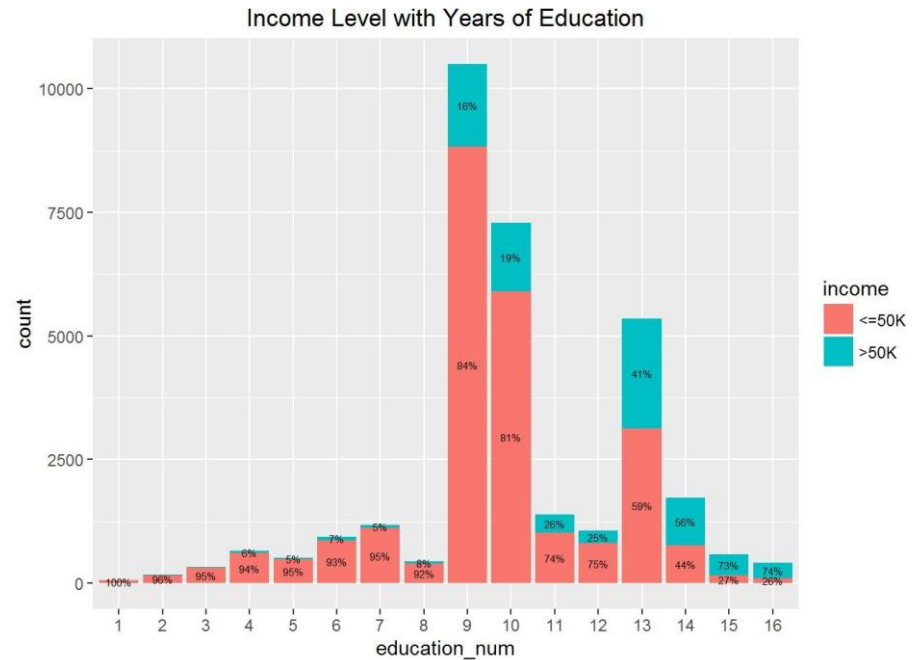
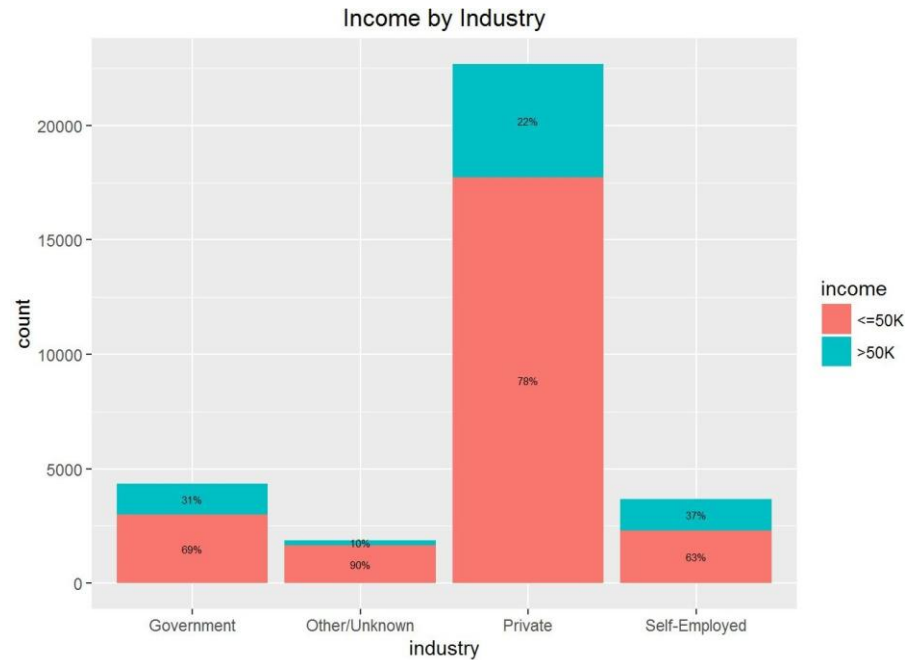


Feature



Feature

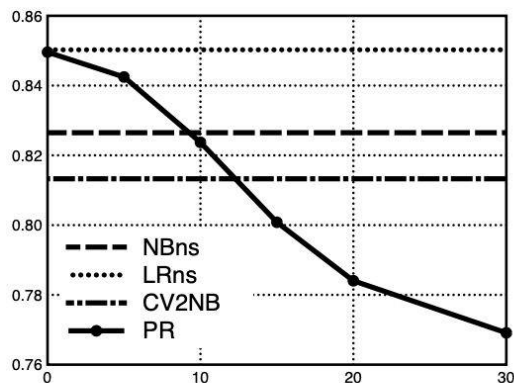
Adult Income Dataset ([Kohavi 1996](#))



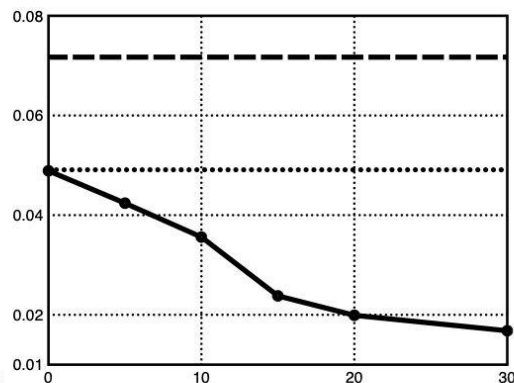
Results



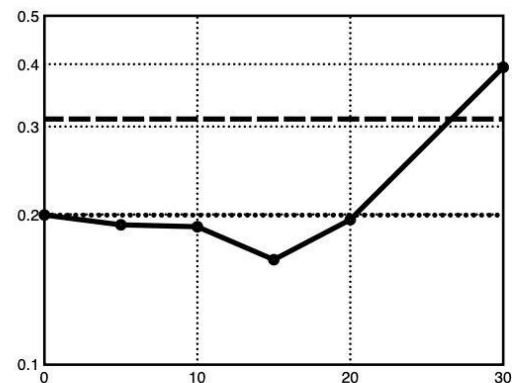
- Changes of Performance With η
 - Model performance decreases (Acc)
 - Discrimination Decreases (NPI)
 - "Fairness Efficiency" (PI/MI) Increases



(a) Acc



(b) NPI



(c) PI/MI

[Kamishima et al, 2012](#)

Results



- Prejudice Prior Sacrifices Model Performance
 - PR has lower Acc (Accuracy)
 - PR has lower NMI (normalized mutual information between labels and predictions)
- Prejudice Prior Makes Model Fair
 - PR has lower NPI

	method	Acc	NMI	NPI	PI/MI
Logistic Regression full fet.	→ LR	0.851	0.267	5.21E-02	2.10E-01
Logistic Regression no sensitive fet.	→ LRns	0.850	0.266	4.91E-02	1.99E-01
	→ PR $\eta=5$	0.842	0.240	4.24E-02	1.91E-01
Logistic Regression + Prejudice Regularizer	→ PR $\eta=15$	0.801	0.158	2.38E-02	1.62E-01
	→ PR $\eta=30$	0.769	0.046	1.68E-02	3.94E-01

η is the weight we put on prejudice regularizers [Kamishima et al, 2012](#)

Results



- PI/MI
 - Prejudice Index / Mutual Information
 - Demonstrates a trade-offs between model fairness and performance
 - Measures the amount of discrimination we eliminate with one unit of performance gain (measured by MI)

		method	Acc	NMI	NPI	PI/MI
Logistic Regression full fet.	→	LR	0.851	0.267	5.21E-02	2.10E-01
Logistic Regression no sensitive fet.	→	LRns	0.850	0.266	4.91E-02	1.99E-01
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	→	PR $\eta=15$	0.801	0.158	2.38E-02	1.62E-01
	↘	PR $\eta=30$	0.769	0.046	1.68E-02	3.94E-01

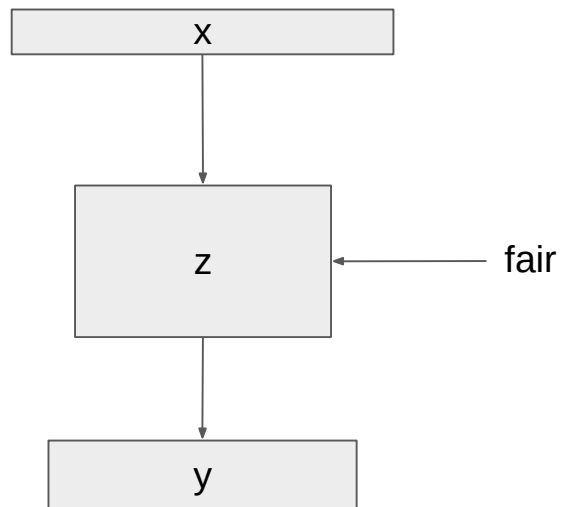
η - weight put on the prejudice regularizer

[Kamishima et al, 2012](#)

Fair Representation Learning



- Ensure fairness up to a certain level

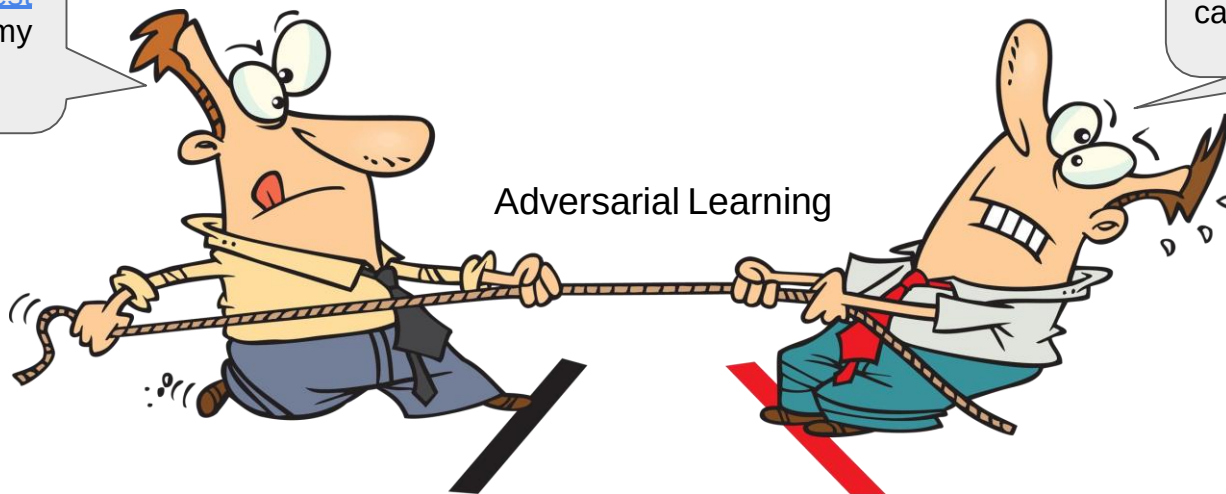


Fair Representation Learning

- How Do We Test the Fairness of Representation Z ?
 - Adversarial Learning

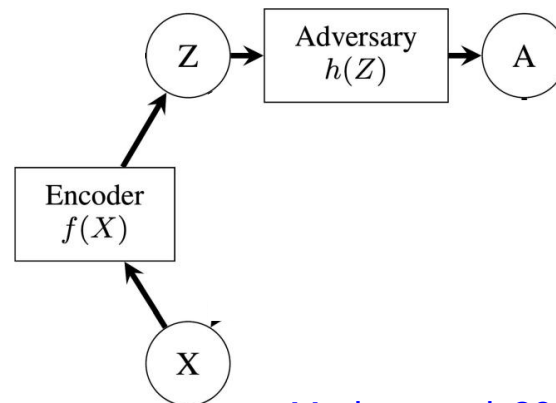
I want to find the best representation for my task.

I want to find the worst representation that can reconstruct A



Fair Representations

- How Do We Make a Representation Z Fair?
 - $Z = f(X)$
 - Test and see if a good amount of A can be reconstructed from Z
 - Compare A with $h(z)$



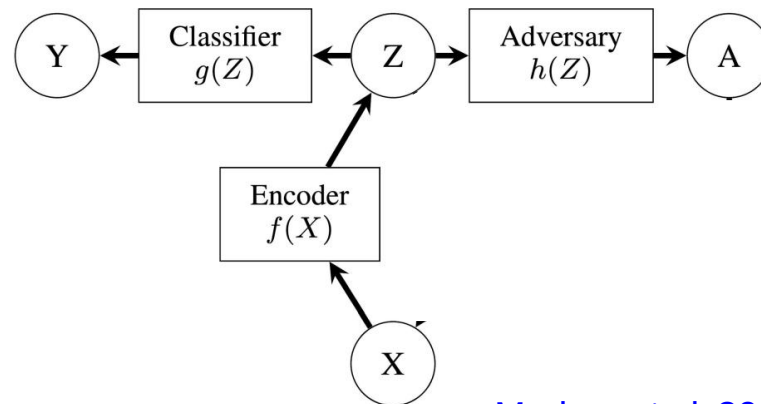
[Madras et al, 2018](#)

Fair Representations



- How Do We Make a Representation Z Fair?
 - $Z = f(X)$
 - Test and see if a good amount of A can be reconstructed from Z
 - Compare A with $h(z)$

- Properties of Deep Representations
 - Achieve good performance for downstream task that generates $y=g(z)$

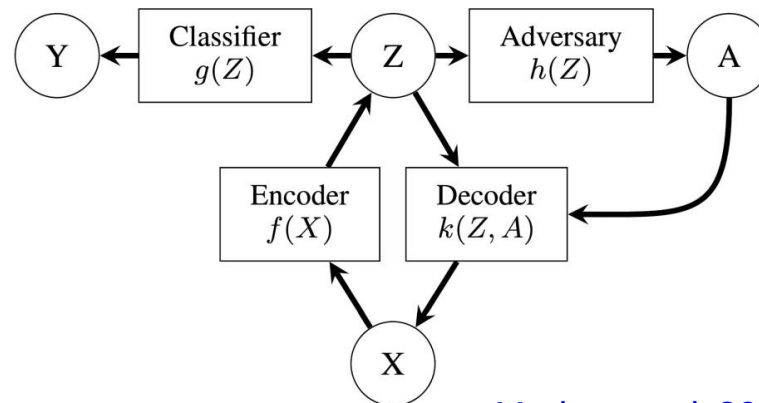


[Madras et al, 2018](#)

Fair Representations

- How Do We Make a Representation Z Fair?
 - $Z = f(X)$
 - Test and see if a good amount of A can be reconstructed from Z
 - Compare A with $h(z)$

- Properties of Deep Representations
 - Achieve good performance for downstream task that generates $y=g(z)$
 - Has the Ability to Reconstruct $X = k(Z, A)$



[Madras et al, 2018](#)

Fairness Through Adversarial Learning

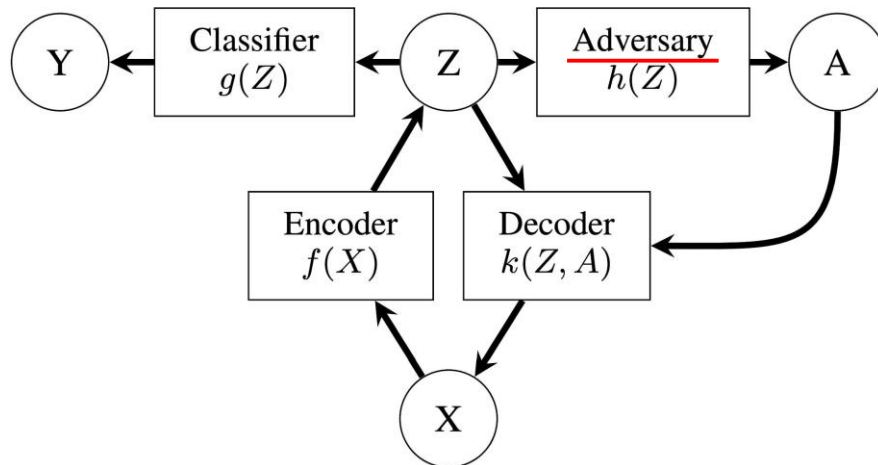
innovate

achieve

lead

- Adversarial Learning

- Models are trained using objectives that compete with each other



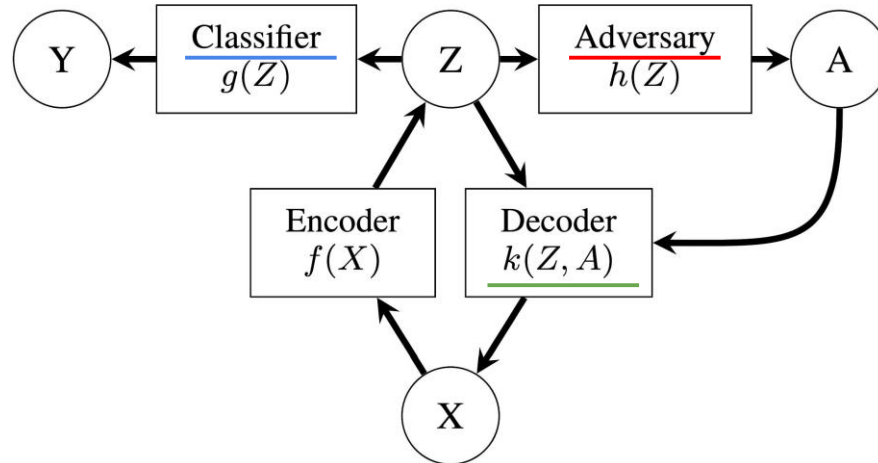
$$\underset{f,g,k}{\text{minimize}} \underset{h}{\text{maximize}} \mathbb{E}_{X,Y,A} [L(f, g, h, k)]$$

[Madras et al, 2018](#)

Fairness Through Adversarial Learning

- Adversarial Learning

$$L(f, g, h, k) = \alpha \underline{L_C(g(f(X, A)), Y)} + \beta \underline{L_{Dec}(k(f(X, A), A), X)} + \gamma \underline{L_{Adv}(h(f(X, A)), A)}$$



$$\underset{f, g, k}{\text{minimize}} \underset{h}{\text{maximize}} \mathbb{E}_{X, Y, A} [L(f, g, h, k)]$$

[Madras et al, 2018](#)

Loss for Learning Fair Representations

- Adversarial Loss for Demographic Parity with Group $\mathcal{D}_0, \mathcal{D}_1$

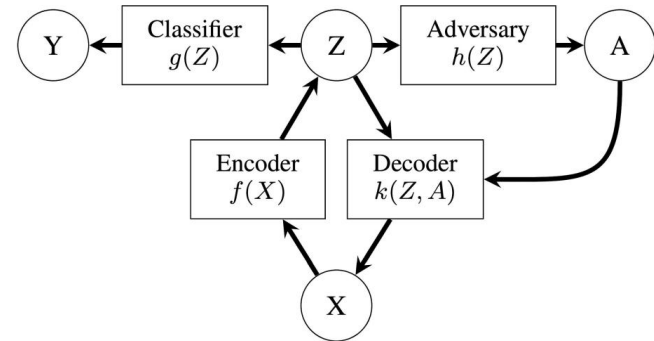
$$L_{Adv}^{DP}(h) = 1 - \sum_{i \in \{0,1\}} \frac{1}{|\mathcal{D}_i|} \sum_{(x,a) \in \mathcal{D}_i} |h(f(x,a)) - a|$$

Demographic Parity: $P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$

- Adversarial Loss for Equality of Odds with Group $\mathcal{D}_i^j = \{(x, y, a) \in \mathcal{D} | a = i, y = j\}$

$$L_{Adv}^{EO}(h) = 2 - \sum_{(i,j) \in \{0,1\}^2} \frac{1}{|\mathcal{D}_i^j|} \sum_{(x,a) \in \mathcal{D}_i^j} |h(f(x,a)) - a|$$

Equality of Odds: $P(\hat{Y} = 1|A = 0, Y) = P(\hat{Y} = 1|A = 1, Y)$



[Madras et al, 2018](#)

Discrimination Measures for Representations

$$\mathcal{Z}_1 = p(Z|A = 1) \quad \mathcal{Z}_0 = p(Z|A = 0) \quad \mathcal{Z}_a^y = p(Z|A = a, Y = y)$$

- Demographic Parity

$$\Delta_{DP}(g) \triangleq d_g(\mathcal{Z}_0, \mathcal{Z}_1) = |\mathbb{E}_{\mathcal{Z}_0}[g] - \mathbb{E}_{\mathcal{Z}_1}[g]|$$

$$\text{Demographic Parity: } P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$$

- Equality of Odds

$$\Delta_{EO}(g) \triangleq |\mathbb{E}_{\mathcal{Z}_0^0}[g] - \mathbb{E}_{\mathcal{Z}_1^0}[g]| + |\mathbb{E}_{\mathcal{Z}_0^1}[1 - g] - \mathbb{E}_{\mathcal{Z}_1^1}[1 - g]|$$

$$\text{Equality of Odds: } P(\hat{Y} = 1|A = 0, Y) = P(\hat{Y} = 1|A = 1, Y)$$

- Equality of Opportunities

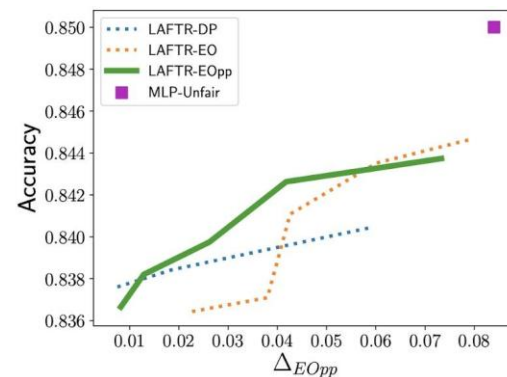
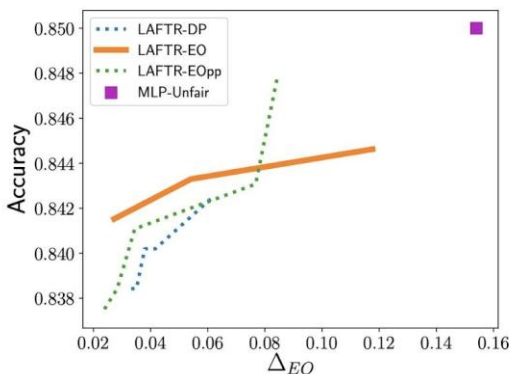
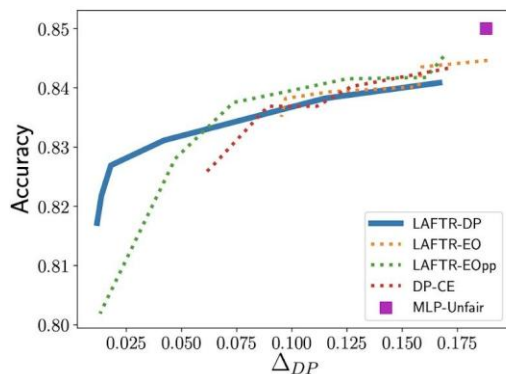
$$\Delta_{EOpp}(g) \triangleq |\mathbb{E}_{\mathcal{Z}_0^0}[g] - \mathbb{E}_{\mathcal{Z}_1^0}[g]|$$

$$\text{Equality of Opportunity: } P(\hat{Y} = 1|A = 0, Y = 1) = P(\hat{Y} = 1|A = 1, Y = 1)$$

Accuracy and Fairness on Adult Income Dataset

- Results Generated By Varying γ

$$L(f, g, h, k) = \alpha L_C(g(f(X, A)), Y) + \beta L_{Dec}(k(f(X, A), A), X) + \gamma L_{Adv}(h(f(X, A)), A)$$



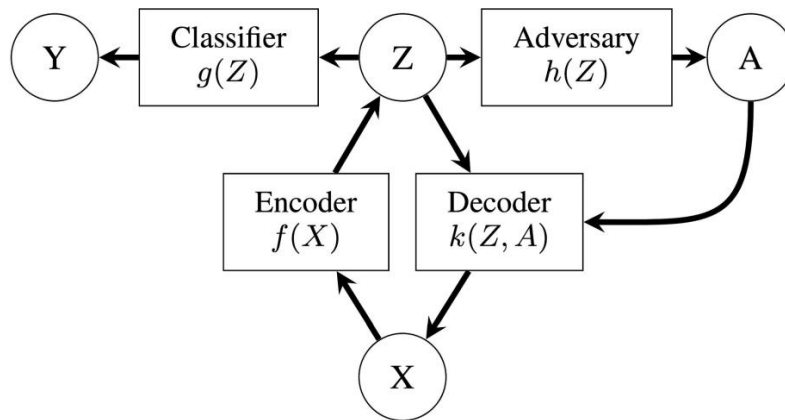
DP-CE - Cross Entropy Adversarial Objective ([Edwards et al, 2016](#))

[Madras et al, 2018](#)

Transferring Fair Representations



- If the Representations Are Fair, All Predictors Should Be Fair!
 - Train f and g based on domain 1 with feature space X
 - Fix f , and train g' on domain 2 with the same feature space X
 - $y=g'(f(x))$ should be a fairness predictor

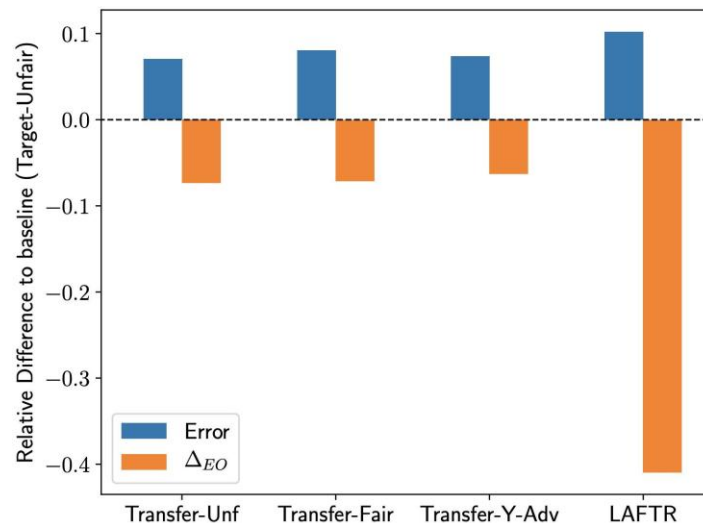


[Madras et al, 2018](#)

Transfer Fair Representations



- Heritage Health Dataset
 - Comprises insurance claims and physician records
 - Task 1 - Predict Charlson index (prediction of 10 year survival of patients) trained using equalized odds adversarial objective
 - Task 2 - Same input, task becomes predicting a patient's insurance claim corresponding to a specific medical condition



Transfer- unf - MLP with no fairness constraints

Transfer- fair - MLP with fairness constraints in [Bechavod et al, 2017](#)

Transfer - Y - Adv baseline in [Zhang et al, 2018](#)

[Madras et al, 2018](#)

In a nutshell

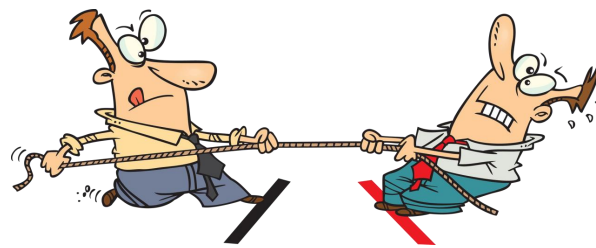
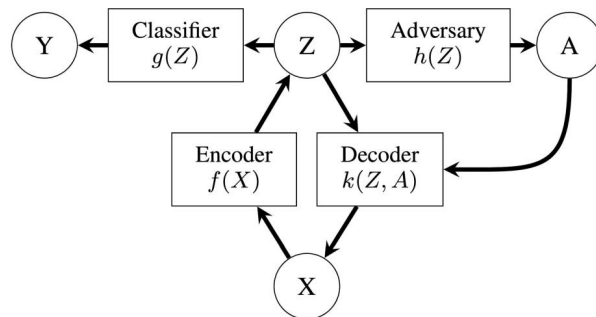


- Fair Representation Learning
 - Prejudice Removing Regularizer

$$-\mathcal{L}(\mathcal{D}; \boldsymbol{\Theta}) + \eta R(\mathcal{D}, \boldsymbol{\Theta}) + \frac{\lambda}{2} \|\boldsymbol{\Theta}\|_2^2$$

Loss of the Model Fairness Regularizer L2 Regularizer

- Fair Representations Through Adversarial Learning



Model Prediction ↑

Reconstruct A ↓

Comparisons: Regularization and Adversarial Learning



	Prejudice Removing Regularizer	Adversarial Learning
Pros	Minimal modifications to training procedure	Transferable representations
		Can be applied to many different fairness criteria
Cons	Can only be applied to Demographic Parity	Adversarial loss can be difficult to train

Readings



- T1, Chapter 3
- <https://www.borealisai.com/research-blogs/tutorial1-bias-and-fairness-ai/>
- Links to the research articles mentioned on the slides